

Overview and Mineral Function:

Calcium is the most abundant mineral in the human body. Over 99% of the body's calcium is stored in the bones and teeth, where it supports structure and hardness. The remaining 1% is found in blood, muscle, and intercellular fluids, playing an indispensable role in vascular contraction, vasodilation, muscle function, nerve transmission, and hormonal secretion.

Recommended Dietary Allowances (RDA) for Calcium:

- Children 1-3 years: 700 mg/day
- Children 4-8 years: 1,000 mg/day
- Adolescents 9-18 years: 1,300 mg/day
- Adults 19-50 years: 1,000 mg/day
- Men 51-70 years: 1,000 mg/day
- Women 51-70 years: 1,200 mg/day
- Adults 71+ years: 1,200 mg/day

Tolerable Upper Intake Levels (UL) for Calcium:

- Children 1-8 years: 2,500 mg/day
- Children 9-18 years: 3,000 mg/day
- Adults 19-50 years: 2,500 mg/day
- Adults 51+ years: 2,000 mg/day

Dietary Sources of Calcium:

Dairy products represent the richest and most bioavailable sources of calcium:

- Milk, yogurt, and cheese (approx. 200-300 mg per serving).
- Fortified Foods: Calcium-fortified orange juices, fortified plant milks (soy, almond, oat), and fortified breakfast cereals.
- Non-Dairy Foods: Calcium-set tofu, canned sardines and salmon with soft edible bones, and dark leafy greens such as kale, bok choy, and broccoli. (Spinach contains calcium, but bioavailability is low due to oxalates).

Bone Health and Osteoporosis:

Adequate calcium intake throughout life, as part of a well-balanced diet, may reduce the risk of osteoporosis and bone fractures in older age. Adequate vitamin D intake is necessary to ensure efficient intestinal calcium absorption. Excess calcium intake (primarily from supplements) may increase the risk of nephrolithiasis (kidney stones).